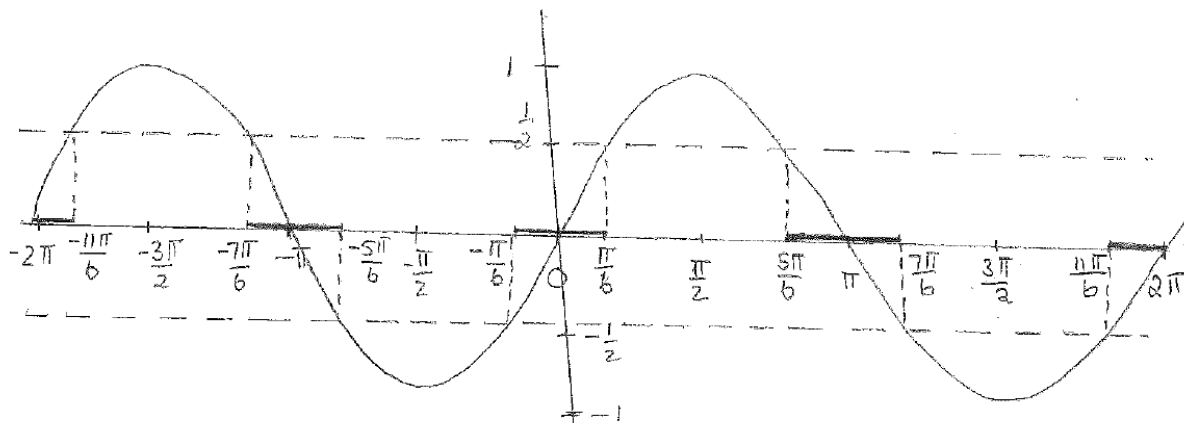


Trig inequality problems

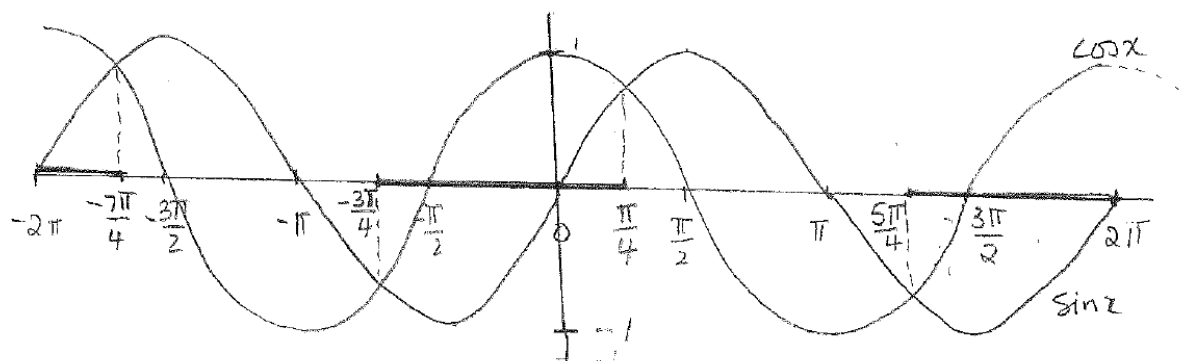
1. Solve $|\sin x| \leq \frac{1}{2}$, $x \in [-2\pi, 2\pi]$

Solution: $|\sin x| \leq \frac{1}{2} \Rightarrow -\frac{1}{2} \leq \sin x \leq \frac{1}{2}$



$x \in [-2\pi, -\frac{11\pi}{6}] \cup [-\frac{7\pi}{6}, -\frac{5\pi}{6}] \cup [-\frac{\pi}{6}, \frac{\pi}{6}] \cup [\frac{5\pi}{6}, \frac{7\pi}{6}] \cup [\frac{11\pi}{6}, 2\pi]$

2. Solve $\cos x > \sin x$, $x \in [-2\pi, 2\pi]$



$x \in [-2\pi, -\frac{7\pi}{4}) \cup (-\frac{3\pi}{4}, \frac{\pi}{4}) \cup (\frac{5\pi}{4}, 2\pi]$

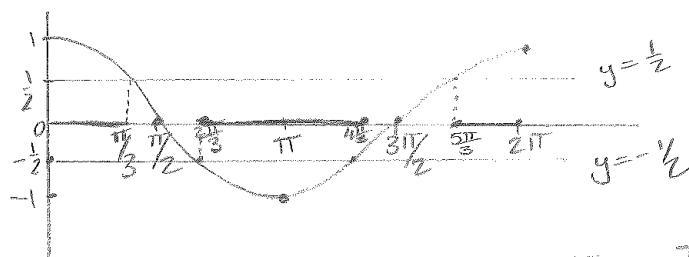
3. Solve: $|\cos x| \geq \frac{1}{2}$, $x \in [0, 2\pi]$

Answer: $x \in [0, \frac{\pi}{3}] \cup [\frac{2\pi}{3}, \frac{4\pi}{3}] \cup [\frac{5\pi}{3}, 2\pi]$

4. Solve: $0 \leq \tan x \leq 1$, $x \in [-\pi, 2\pi]$

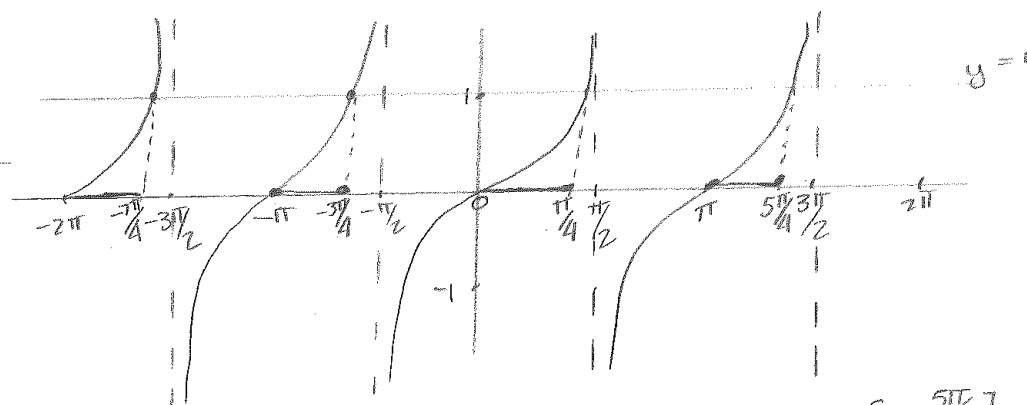
Answer: $x \in [-\pi, -\frac{3\pi}{4}] \cup [0, \frac{\pi}{4}] \cup [\pi, \frac{5\pi}{4}]$

3) Solve $|\cos x| \geq \frac{1}{2}$, $x \in [0, 2\pi]$ $\Rightarrow \cos x \geq \frac{1}{2}$ or $\cos x \leq -\frac{1}{2}$



$$x \in [0, \frac{\pi}{3}] \cup [\frac{2\pi}{3}, \frac{4\pi}{3}] \cup [\frac{5\pi}{3}, 2\pi]$$

4) Solve $0 \leq \tan x \leq 1$, $x \in [-2\pi, 2\pi]$



$$x \in [-2\pi, -\frac{7\pi}{4}] \cup [-\pi, -\frac{3\pi}{4}] \cup [0, \frac{\pi}{4}] \cup [\pi, \frac{5\pi}{4}]$$